BME 423: Statistical and Data Science Methods in Biomedical Engineering
Fall 2026

1. Basic Information

Course:	Statistical & Data Science Methods in Biomedical Eng., BME 423, 4 units
Lecture:	Section 29250: M&W: 10:00 to 11:50 AM, GFS 116
Discussion:	Section 29256: Tu, 4:00 to 5:20 PM (TA: Zixin), SLH 102
	Section 29257: Th, 12:30 to 1:50 PM (TA: Zixin), GFS 118
Faculty:	David Z. D'Argenio, Ph.D., Professor, BME (DRB 154)
Email:	dargenio@usc.edu **(put "BME 423" in subject line)**
Office Hours:	W, 1:00 to 2:30 PM. DRB 154
TAs:	Zixin Chen (chenzixi@usc.edu)
TA Office Hours:	Zixin Chen, Tu, 5:30 to 7:30 PM, via Zoom
	https://usc.zoom.us/j/93598373417
Graders:	Vedika Kothari (vedikako@usc.edu)
	Cooper Segall (csegall@usc.edu)
Final Exam:	Monday, December 15, 8:00 to 9:30 AM
Prerequisite:	MATH 245
Class webpage:	https://brightspace.usc.edu/d2l/home (log in and select BME 423)

2. Course Goals, Learning Objectives, Relationship to Program Outcomes & BME Courses

2.1 Goals

The goal of BME 423 is to introduce statistical and data science methods for making inferences in engineering, biology, and medicine. You will learn fundamental techniques for analyzing data arising from different types of biomedical experiments and clinical trials. You will learn about: descriptive statistics; elementary probability; discrete and continuous random variables and their distributions; hypothesis testing involving interval (continuous and discrete) and categorical (nominal and ordinal) variables, for two and three-or-more treatments; simple and multiple linear regression; logistic regression; machine learning for clustering, classification and prediction; analysis of time-to-event (survival) data; clinical trials. You will also be introduced to the R software environment for statistical data analysis. The overall objective of this course is to help you learn how to make decisions under uncertainty.

2.2 Course learning objectives

After successfully completing this course, you should be able to:

- **Objective 1:** explain and apply descriptive statistics (including measures of central tendency and dispersion) to summarize experiment data
- **Objective 2:** know what a random variable is (discrete and continuous), some common probability distributions, and how to calculate expected values
- **Objective 3:** explain and implement methods for testing differences between treatment groups when the measurements are interval-type data, including: analysis of variance; unpaired and paired t test; repeated measures ANOVA
- **Objective 4:** explain and implement methods for testing differences between treatment groups when the measurements are categorical, including: Chi-squared analysis; Mann Whitney test; Kruskal-Wallis statistic; Wilcoxon signed-rank test; McNemar's test
- **Objective 5:** describe and implement methods for assessing the relation between two or more variables using linear regression analysis and logistic regression
- **Objective 6:** explain and implement machine learning algorithms for clustering, classification and prediction
- **Objective 7:** explain and implement analysis of time-to-event (survival) data and tests for evaluating the effects of treatments using time-to-event data
- **Objective 8:** select and implement the appropriate statistical procedure needed for a different experiment paradigm, data collected, and question to be answered
- **Objective 9:** design a clinical trial to test whether a treatment produces an effect (type of experiment, number of subjects, etc.), considering ethical issues of experiments using humans
- **Objective 10:** prepare written reports that document solutions to homework projects and present results and analyses
- **Objective 11:** supplement through independent study of reference readings, the statistical concepts presented in class and their biomedical applications

2.3 Relation of course learning objectives to BME Program Outcomes (listed in Section 11 below)

Student Outcomes → **↓ Course Objectives**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Objective 1	x					x	
Objective 2	x					x	
Objective 3	x					x	
Objective 4	x					x	
Objective 5	x	x				x	
Objective 6	x	x				x	
Objective 7	x	x				x	
Objective 8	x	x				x	
Objective 9		x		x		x	
Objective 10			x				
Objective 11							x
All course objectives	x	x	x	x		x	x

2.4 Relation of BME 423 to the other BME courses

At this point in your studies, most of you have taken the following BME courses: BME 101 (Introduction to Biomedical Engineering), BME 202 (Control and Communication in the Nervous System), and BME 210 (Biomedical Computer Simulation Methods). An overview of the material you covered in these courses is provided below (Section 12 of this syllabus), which summarizes the concepts, physiological processes, and methods introduced in these courses. The corresponding summary is also shown for BME 423. These high-level synopses are provided to guide you in understanding the overall connections between these courses and also to those BME courses that you will take in the future.

2.5 Course Organization

The lecture and discussion outlines presented below reflect the course goals and program learning objectives. The lectures will emphasize the different statistical and data science methods, while the discussion section reviews and applies these concepts, and introduces software to implement the methods and procedures presented in the lectures. The discussions will also provide an opportunity for you to ask further questions regarding the lecture material and homework assignments, and to review assignment solutions. The homework projects require an understanding of when to use specific statistical tests and analysis procedures given the experiment data and question posed, as well as how to perform the procedure.

3. Schedule of Lectures

Week	Session (Day)	Date	Topics	Reading/Homework
1	1 (M)	8/24	Introduction	*Glantz, Ch.1, p. 1-5*
1	2 (W)	8/26	Descriptive Statistics	*Glantz, Ch.2, p. 7-26*
2	3 (M)	8/31	Probability/Random Variables	*Course slides* **HW#1 assigned**
2	4 (W)	9/2	Probability/Random Variables (Exercise)	*Course slides*
3	**(M)**	**9/7**	***Labor Day Holiday***	**--**
3	5 (W)	9/9	Probability/Random Variables	*Course slides*
4	6 (M)	9/14	Differences Among Groups	*Glantz, Ch.3, p. 27-40*
4	7 (W)	9/16	Differences Among Groups (Exercise)	*Glantz, Ch.3, p. 41-47*
5	8 (M)	9/21	Special Case of Two Groups	*Glantz, Ch.4, p. 49-60*
5	9 (W)	9/23	Special Case of Two Groups (Exercise)	*Glantz, Ch.4, p. 61-72*
5	**(F)**	**9/25**	**--**	**Homework #1 due at 5 PM, via Brightspace**
6	10 (M)	9/28	Sampling Distributions Clinical Trials	*Course slides* *Glantz, Ch.12, p.247-258* **HW#2 assigned**
6	11 (W)	9/30	Proportions & Nominal Variables (Exercise)	*Glantz, Ch.5, p. 73-80*
7	12 (M)	10/5	Proportions & Nominal Variables	*Glantz, Ch.5, p. 81-90*
7	13 (W)	10/7	Measures of Association: Nominal Variables (Exercise)	*Glantz, Ch.5, p.91-95*
7	**(Th-F)**	**10/8-10/9**	***Fall Recess***	**--**
8	14 (M)	10/12	Types of Errors in Inference & Power of a Statistical Test	*Glantz, Ch.6, p.101-124*
8	15 (W)	10/14	Midterm Exam Review	--
8	**(F)**	**10/16**	**--**	**Homework #2 due at 5 PM, via Brightspace (hard deadline so solutions can be posted)**
9	**16 (M)**	**10/19**	**MIDTERM EXAM (90 minutes)**	**--**
9	17 (W)	10/21	Confidence Intervals (Exercise)	*Glantz, Ch.7, p.125-142* **HW#3 assigned**
10	18 (M)	10/26	(Simple) Linear Regression	*Glantz, Ch.8, p143-178*
10	19 (W)	10/28	Multiple Linear Regression & Logistic Regression (Exercise)	*Course slides*
11	20 (M)	11/2	Multiple Treatments (Exercise)	*Glantz, Ch.9, p.185-194*
11	21 (W)	11/4	ANOVA for Multiple Treatments & McNemar's Test	*Glantz, Ch.9, p.195-202*
12	22 (M)	11/9	Machine Learning: Clustering (Exercise)	*Course slides*
12	**(W)**	**11/11**	***Veterans Day Holiday***	
12	**(F)**	**11/13**	**--**	**Homework #3 due at 5 PM, via Brightspace**
13	23 (M)	11/16	Machine Learning: Decision Trees (Exercise)	*Course slides* **HW#4 assigned**
13	24 (W)	11/18	Nonparametric Methods (Exercise)	*Glantz, Ch.10, p.205-216*
14	25 (M)	11/23	Nonparametric Methods (Exercise)	*Glantz, Ch.10, p.217-227*
14	**(W-Su)**	**11/25-11/29**	***Thanksgiving Recess***	**--**
15	26 (M)	11/30	Time-to-Event (Survival) Analysis	*Glantz, Ch.11, p.229-244*
15	27 (W)	12/2	Final Exam Review	--
15	**(F)**	**12/4**	**--**	**Homework #4 due at 5 PM, via Brightspace**
FINAL EXAM	**(M)**	**12/14**	**FINAL EXAM 8:00 AM (90 minutes)**	GFS 116

4. Discussion Sessions

The overall goals of the Discussion Sessions are to 1) help you to better understand the concepts and procedures presented in the lectures, and 2) guide you in applying the R software to perform the analyses/statistical tests/calculations. For the latter, you will also need to gain some familiarity with R so that you can perform the basic data manipulation and plotting needed to implement and interpret R's statistical test and analysis functions. We emphasize, however, that our goal is not to introduce all aspects of the R programming language, but to give you enough basic familiarity with R so that you can manipulate data and implement the statistical tests and analysis methods introduced in the lectures.

The Discussion Sessions will comprise the following types of activities:

1. Questions: At the beginning of each session, you will have the opportunity to ask questions related to the material covered in the previous lectures and discussion sessions, as well as on the current homework assignment.
2. Homework Assignment Overview: The homework assignments will be presented (see Lecture Schedule above for assignment and due dates).
3. Guided Tour: Interactive walk-through of R commands/code needed to implement a statistical test function, analysis, or data display.
4. Problem Solving Exercise: You will be given a task/problem to complete. The solutions will then be presented and discussed as a group. These exercises may involve by-hand implementation of a statistical test introduced in the lectures and/or will require solutions using the R software.
5. Discussion Log: For most discussion sessions, you will be asked to submit a log of work you have done during that session (e.g., a log of all of your inputs to R and all of the outputs from R).

Discussion Schedule

Week	Discussion Topics
1	Install and set up R and RStudio (No Log)
2	Learning to use R (Log) Data manipulation Plotting with R
3	Descriptive statistics (Log)
4	Probability calculations (Log) ANOVA: **aov()** function
5	Homework 1 Q&A (No Log)

6	Versions of the t Test: **independentSamplesTTest()**, **oneSampleTTest()**, and **posthocPairwiseT()** functions (Log)
7	NO DISCUSSION (Fall Recess)
8	χ^2 Test: **chisq.test()** function (Log) Homework 2 Q&A
9	Types of Errors/Power of a Test (Log)
10	Linear regression: **lm()** function (Log) Correlation: **cor()** function
11	Multiple linear regression: **lm()** function (Log) Logistic regression: **glmfit()** function Homework 3 Q&A
12	NO DISCUSSION (Veterans Day Holiday)
13	Clustering: **kmeans()** function (Log) Classification: **knn()** function Prediction: **neuralnet()** function
14	NO DISCUSSION (Thanksgiving Recess)
15	Survival analysis: **surv()**, **survfit()**, and **survdiff()** functions (Log) Homework 4 Q&A

5. Required Text and Source Materials

1. Primer of Biostatistics. Stanton A. Glantz, McGraw Hill, 7th edition, 2012. (CD not needed)
   This textbook can be accessed for free using your USC login as part of USC Library's subscription to McGraw Hill's "Access Science"
   On USC's network: https://www.accessscience.com/content/book/9780071781503
   Off campus: https://libproxy.usc.edu/login?url=https://www.accessscience.com/

2. BME 423 Lecture/Discussion Slides. Download from BME 423 course website on Brightspace

3. R for Beginners. Emmanuel Paradis. PDF is free to download from https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf

4. R (free to download from https://www.r-project.org/) and RStudio Software (free to download from https://www.rstudio.com/)

6. Grading

Midterm Exam	25%
Final Exam	35%
Homework (4)	20%
Lecture Exercises (10)	10%
Discussion Logs (8)	10%

The homework will consist of four written assignments. All assignments are to be turned in (as a PDF uploaded to Brightspace) as indicated on the homework prompt. For every day (or portion of a day) an assignment is late, 25% will be subtracted from its maximum point total.

While you may consult with classmates regarding the homework assignments and on solution ideas, you are required to write your own homework reports. Homework solutions may not be shared. Plagiarism of another's work (including that of your classmates) is a serious offense and all suspected cases will be handled according to University regulations (see SCampus: http://scampus.usc.edu/).

Lecture exercises will consist of the solution to a problem presented in class. You can work in groups of two or three and will be given 10 or 15 minutes to complete your solution. The exercise solution will be submitted via Brightspace during class as a photo or PDF of your written work. Each student in a group can submit the same solution as the other group members. Twelve Lecture Exercises will be assigned, and you will be graded on any 10 that you submit. Grading will be on a credit/no-credit basis (no partial credit) and there will be no extensions or makeups.

Discussion logs will consist of the inputs and outputs to/from R during a given discussion session. Each log will be due at the end of the week containing the associated discussion session (no extensions or late submissions). Logs will be submitted via Brightspace as a PDF, Word document, or plain text file. Ten Discussion Logs will be assigned, and you will be graded on any eight that you submit. Grading will be on a credit/no-credit basis (no partial credit).

For both exams, students will be allowed to bring a calculator and two double-sided, 8.5" by 11" note pages. These note pages may either be typed/printed-out, handwritten, or a combination of both.

7. Web Page

This course uses Brightspace and can be accessed via https://brightspace.usc.edu/d2l/home.

8. Academic Conduct

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct is in contrast to the university's mission to educate students through a broad array of first-rank academic, professional, and extracurricular programs and includes any

act of dishonesty in the submission of academic work (either in draft or final form).

This course will follow the expectations for academic integrity as stated in the USC Student Handbook. All students are expected to submit assignments that are original work and prepared specifically for the course/section in this academic term. You may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of engaging in academic misconduct will be reported to the Office of Academic Integrity.

Other violations of academic misconduct include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

Plagiarism – presenting someone else's ideas as your own, either verbatim or recast in your own words – is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, "Behavior Violating University Standards" policy.usc.edu/scampus-part-b. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on scientific misconduct, policy.usc.edu/scientific-misconduct.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Use of Generative AI in this Course

Generative AI is permitted but limited as follows:

In this course, you are permitted to use artificial intelligence (AI) tools.

- You should be aware that AI text and code generation tools may present incorrect information, biased responses, and incomplete analyses; thus, their answers may not meet the standards of this course.
- To adhere to our values at USC, you must cite any AI-generated material (e.g., text, images, and other content) included or referenced in your work and provide the prompts used to generate the content. Using an AI tool to generate content without proper attribution will be treated as plagiarism and reported to the Office of Academic Integrity.

9. Students and Disability Accommodations

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

10. Support Systems

Covid-19 Related Information

- Classrooms: https://ehs.usc.edu/files/covid19-classroom-best-practices-v2.2.pdf;
- Food and Drink: https://ehs.usc.edu/files/covid19-food-drink-best-practices-v2.2.pdf;
- Workplace: https://ehs.usc.edu/files/covid19-office-best-practices-v2.pdf

Counseling and Mental Health – (213) 740-9355 – 24/7 on call
studenthealth.usc.edu/counseling
Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline – 988 for both calls and text messages – 24/7 on call
https://988lifeline.org/
The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

Relationship and Sexual Violence Prevention Services (RSVP) – (213) 740-9355(WELL), press "0" after hours – 24/7 on call
studenthealth.usc.edu/sexual-assault
Free and confidential therapy services, workshops, and training for situations related to gender-based harm.

Office for Equity, Equal Opportunity, and Title IX (EEO-TIX) – (213) 740-5086
https://eeotix.usc.edu/
Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

Reporting Incidents of Bias or Harassment – (213) 740-5086 or (213) 821-8298
usc-advocate.symplicity.com/care_report
Avenue to report incidents of bias, hate crimes, and microaggressions to the Office of Equity and Diversity |Title IX for appropriate investigation, supportive measures, and response.

The Office of Student Accessibility Services (OSAS) – (213) 740-0776
https://osas.usc.edu/
OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention – (213) 821-4710
campussupport.usc.edu
Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Diversity, Equity, and Inclusion – (213) 740-2101
diversity.usc.edu
Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency – UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call
dps.usc.edu, emergency.usc.edu
Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety – UPC: (213) 740-6000, HSC: (323) 442-120 – 24/7 on call
dps.usc.edu
Non-emergency assistance or information.

Office of the Ombuds – (213) 821-9556 (UPC) / (323-442-0382 (HSC)
ombuds.usc.edu
A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice – (323) 442-2850 or otfp@med.usc.edu
chan.usc.edu/patient-care/faculty-practice
Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

11. Student Outcomes of Biomedical Engineering Program

Students completing the program have

1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factor

3) an ability to communicate effectively with a range of audiences

4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

12. BME Undergraduate Course Overview

The individual courses that you will take as students in biomedical engineering are each components of an integrated curriculum of study. The summary below, which details the core concepts, biomedical processes and tools that you will be introduced to in each course, should serve as a roadmap to help you recognize and further explore the connections among these courses.

Required for All Programs

BME 101 Introduction to Biomedical Engineering (4)

Concepts/Principles
- Conservation of mass
- Conservation of charge
- Electrical circuit theory

Physiological/Cellular/Molecular Processes
- Cellular reactions: cellular respiration, production of biomass
- Membrane potentials
- Renal filtration (also dialysis)

Methods/Tools
- Excel (basic)
- MATLAB
- Arduino (basic circuit wiring, some measurements)

BME 202 Control & Communication in the Nervous System (4)

Concepts/Principles
- Ion channel transport
- Electrical circuit theory
- Cable theory
- Kinetic models
- Poisson process
- Oscillatory circuits

Physiological/Cellular/Molecular Processes
- Action potential propagation
- Synaptic transmission
- Sensory systems (visual, auditory, somatosensory)
- Motor system

Methods/Tools
- Ordinary differential equations
- Image filtering
- Population decoding
- MATLAB, Excel

BME 210 Biomedical Computer Simulation Methods (4)

Concepts/Principles

- Mass action kinetics
- Conservation of mass
- Mass transport
- Electromagnetic absorption/emission for imaging

Physiological/Cellular/Molecular Processes

- Receptor-ligand interactions
- Circulatory system (measuring cardiac output)
- Material exchange in tissues (tracer, pharmaco- and toxico-kinetics)
- X-ray computed tomography (image reconstruction)
- Spread of disease

Methods/Tools

- Ordinary differential equations
- Partial differential equations
- Large scale linear algebraic systems
- Stochastic simulations
- MATLAB

BME 403L Physiological Systems (4)

Concepts/Principles

- Electrical Circuits
- Linear Control Theory
- Mechanical Circuits
- Statics
- Dynamics

Physiological/Cellular/Molecular Processes

- Respiration
- Muscle
- Circulation
- Renal

Methods/tools

- MATLAB
- Nonlinear Differential Equations

BME 405L Senior Projects: Measurement & Instrumentation (4)

Concepts/principles

Engineering design process
Time/frequency domain signal analysis – Aliasing
Sensor calibration (static and dynamic)
State machines – Event programming - Design patterns

Physiological/Cellular/Molecular Processes

Varied – context for senior project

Methods/tools

LabVIEW programming
Data acquisition
Microcontroller programming (Arduino – RIO)
Wireless communication (Bluetooth)
Characterization and use of sensors and transducers
CAD design (SolidWorks)
3-D printing - soldering – machining - benchtop development of prototypes

BME 413 Bioengineering Signals and Systems (4)

Concepts/Principles

Signal Analysis
Signal Transforms
Linear Systems Analysis

Physiological/Cellular/Molecular Processes

Systems Biology modeling
Imaging Systems modeling
Image Analysis and Processing

Methods/Tools

Fourier and Laplace Transforms
Convolution
MATLAB

BME 416L Development and Regulation of Medical Devices (4) or BME 415 Regulation of Medical Products (2)

Concepts/Principles

- Regulatory science
- Quality systems
- Risk
- Design controls
- Ergonomics
- Reliability
- Process validation
- Intellectual property law
- Business economics

Physiological/Cellular/Molecular Processes

- Systems physiology
- Foreign body response

Methods/Tools

- Risk analysis
- Project management
- Biocompatibility testing
- Clinical trial design and power analysis
- Patent search, business valuation, and accounting

BME 423 Statistical Methods in Biomedical Engineering (4)

Concepts/principles

- Probability (introduction)
- Descriptive data analysis
- Statistical inference
- Experiment design
- Ethical issues in clinical trials

Physiological/Cellular/Molecular Processes

- Methods applied to experiments/clinical trials involving different biomedical processes

Methods/tools

- Inferential methods for interval data (ANOVA, t test)
- Inferential methods for nominal data (Chi squared, relative risk, odds ratio)
- Nonparametric inferential methods (Wilcoxon, Kruskal-Wallis)
- Time-to-event data analysis (survival analysis)
- Inferential methods for multivariate associations,
- Power of a test, design of experiment, number needed to treat
- R software for statistical analysis

Required in Program Tracks

BME 302L Medical Electronics (4) - Required for BME

Concepts/principles

- Circuit laws - conservation of charge
- Signal shaping: amplification – filtering - rectification
- Device models vs actual devices – tolerance
- Reading engineering documents

Physiological/Cellular/Molecular Processes

- Electromyography
- Electrocardiography
- Blood pressure
- (Seen in lab as context for experimental projects)

Methods/tools

- Circuit analysis and design
- Breadboarding - debugging
- Circuit simulation software (SPICE-based)
- Filter design techniques and software
- Benchtop measurements and tools (scope, DMM, waveform generator)
- Electronic devices: diodes, op-amps, instrumentation amps, voltage regulators
- Microcontroller programming (Arduino)
- Datasheets and app notes

BME 404 Orthopedic Biomechanics (4) - Required for BMME, Anchor course option for BME

Concepts/Principles

- Statics
- Strength of Materials; Stress and Strain Analysis
- Orthopaedic Biomaterials
- Orthopaedic Implant Design Concepts

Physiological/Cellular/Molecular Processes

- Functional Anatomy (Muscles, Tendons, Ligaments)
- Anatomy/physiology of bones & joints (fracture healing)

Methods/tools

- Mechanical Design
- Laboratory modeling using full scale loads and motions
- Instrumentation for strain analysis
- Micromotion measurements

BME 406 Introduction to Bioengineering in Medicine (4) Panel option in BMCE

Concepts/Principles

- Physiology, pathology
- Drug discovery, clinical trials
- Microfabrication, nanotechnologies
- Microscopy
- Surface chemistry
- Animal models

Physiological/Cellular/Molecular Processes

- Epithelial morphogenesis
- Carcinogenesis
- Cancer progression
- Tumor microenvironment
- Cancer migration and metastasis
- Cancer immunology
- Immunotherapy

Methods/Tools

- Micro/nanofabrication
- Microscopy
- Flow cytometry
- Single-cell techniques
- Immunobioengineering

BME 410 Intro. to Biomaterials and Tissue Engineering (4) - Required for BME, Panel option in BMCE

Concepts/Principles

- Stem cells
- Genome editing
- Biomaterials
- Microfabrication/microfluidics
- Tissue engineering

Physiological/Cellular/Molecular Processes

- Stem cell differentiation
- Cell-cell and cell-matrix interactions
- Cardiovascular system

Methods/Tools

- Reading and critiquing of journal articles

BME 430 Principles and Applications of Systems Biology (4) – Panel option in BMCE, Anchor course option for BME

Concepts/Principles

- Mass action kinetics
- Michaelis-Menten kinetics
- Kinetic models
- Analysis of dynamical systems
- Graph theory

Physiological/Cellular/Molecular Processes

- Receptor-ligand interactions
- Metabolic networks
- Signaling pathways
- Gene regulation

Methods/Tools

- Ordinary differential equations
- Linear Algebra
- MATLAB

BME 451 Fundamentals of Biomedical Microdevices (4) Anchor course option for BME

Concepts/Principles

- Physics of miniaturization
- Material properties and crystallography
- Fabrication physics and processes
- Mass transport
- Sensors and sensing principles
- Engineering design process
- Reading journal papers

Physiological/Cellular/Molecular Processes

- Varied - a whole range is covered in the context of different device types

Methods/Tools

- Device design and layout
- Fabrication methods and tools
- Technical presentation and writing
- Simple laboratory techniques

BME 459L Nanomedicine and Drug Delivery (4) Panel option in BMCE

Concepts/Principles

- Transport Phenomenon/diffusion
- Experimental design
- Mechanical properties
- Polymers and degradation
- Grant writing

Physiological/Cellular/Molecular Processes

- Receptor-ligand interactions
- Active and passive targeting
- Blood brain barrier
- Cancer Biology
- Biological transport (active/passive)
- Pharmacokinetics: clearance, half-life, biodistribution, administration route
- Circulatory system
- Toxicity

Methods/Tools

- electron microscopy
- dynamic light scattering
- in vivo half life
- histology
- in vitro biocompatibility